Consulting Project Report

Understanding the compensation policy impact on local resident's tolerance towards wild boars in Giant Panda National Park, China

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Abstract

Compensation payments are widely used to offset the costs of human–wildlife conflict (König et al., 2020, Madden, 2004) and, in doing so, are assumed to raise local tolerance toward the species involved (Hemminger et al., 2025, Kichloo et al., 2023). We test this assumption for wild boar in a household survey of 625 residents across two counties bordering Giant Panda National Park, Sichuan Province, China (Section 2), using a tolerance measure based on the Wildlife Tolerance Model (Kansky et al., 2016) and partial least squares structural equation modelling (Ray et al., 2026). Contrary to the policy assumption, compensation recipients report significantly *lower*, not higher, tolerance than non-recipients (Section 4.3) – a "compensation paradox" that persists after accounting for a strong confound with county of residence (Section 2.2), that is not explained by any of five candidate psychological mechanisms we examined (mediation tests following Nitzl et al., 2016), and that holds up under extensive robustness testing across thousands of alternative analytical choices (specification-vibration analysis following Patel et al., 2015) and across compensation groups (Section 5). Across every check, residents' intangible costs – their attitudes toward and emotional response to wild boar – consistently outweigh compensation status as the strongest correlate of tolerance, in line with earlier work showing that intangible factors shape tolerance (Kansky and Knight, 2014, Kansky and Kidd, 2024). We find no evidence that receiving compensation goes together with the higher tolerance the policy is meant to produce, and the pattern suggests that coexistence-oriented policy would benefit from addressing these intangible costs directly. Because the survey is cross-sectional and compensation was not randomly assigned, these are associations, not causal effects. We recommend a longitudinal, before/after study design as the necessary next step to establish whether this association reflects a causal effect of compensation itself (Section 6.4).

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1 Introduction

Human–wildlife conflict (HWC) – interactions between humans and wildlife that produce negative outcomes such as crop damage, livestock loss, and threats to human safety – is one of the most pressing challenges in contemporary wildlife conservation (König et al., 2020, Madden, 2004, Zimmermann et al., 2020). Beyond its ecological consequences, HWC is fundamentally a social issue: whether it escalates or subsides depends heavily on how the people who bear its costs come to feel about the wildlife causing them. Within HWC research, *tolerance* – the ability and willingness of residents to absorb the real or perceived costs of living alongside wildlife – has emerged as the key indicator of whether coexistence is achievable (Expósito-Granados et al., 2019, Kansky and Kidd, 2024). Tolerance is shaped not only by tangible costs and benefits such as economic losses or compensation payments, but also by intangible dimensions – emotional responses, perceived risk, and the social meanings people attach to wildlife (Kansky and Knight, 2014).

Compensation schemes, which reimburse residents for wildlife-caused damage, are a widely used policy instrument intended to offset these costs and, in doing so, raise tolerance (Hemminger et al., 2025, Kichloo et al., 2023). In China, wild boar (*Sus scrofa*) populations have expanded rapidly in recent decades, and with them the frequency of crop damage and compensation claims in and around national parks. Whether these compensation payments actually achieve their intended social effect – higher tolerance among the people who receive them – is the open empirical question this project addresses. Preliminary evidence from our project partner's fieldwork suggested the opposite of the policy assumption: compensation recipients did not show higher tolerance than non-recipients, and if anything the association ran the other way. Understanding whether that pattern holds up under careful statistical scrutiny, and if so, why compensation might fail to translate into tolerance, is central to designing socially sustainable wildlife-management policy.

This report documents a statistical consulting collaboration between the Department of Statistics at Ludwig-Maximilians-Universität München and project partner Mengxi Kou (Rachel Carson Center for Environment and Society), built around household survey data from two counties bordering Giant Panda National Park, Sichuan Province, China. The analysis is organized around three research questions, unchanged from the original project proposal:

RQ1 Does compensation for wild boar damage have a direct effect on tolerance toward wild boar, or is its influence mediated by other social and psychological factors?

RQ2 What roles do tangible costs, intangible costs, and intangible benefits play in shaping tolerance, and how do these factors interact with compensation experience?

RQ3 How robust are these relationships under different model specifications and across groups with and without compensation experience?

Section 2 describes the survey and the data-quality checks that shaped every later modelling decision, most importantly a strong confound between compensation status and county of residence. Section 3 covers the measurement work: which survey items reliably

capture the constructs the original Wildlife Tolerance Model (Kansky et al., 2016) assumes, and which do not. Section 4 presents the main structural model and the findings for RQ1 and RQ2. Section 5 subjects those findings to two independent robustness exercises – a specification-vibration analysis and a permutation-based multi-group comparison – addressing RQ3. Section 6 discusses what the findings do and do not support, and what would be needed to strengthen the next round of data collection. Section 7 summarizes the project's conclusions.

A note on terminology: throughout this report we use short construct codes (e.g. *IC* for Intangible Cost, *CE0* for compensation participation) consistently with the analysis scripts and the presentation delivered to the project partner, because these codes make cross-referencing tables and models unambiguous. Every code is spelled out in full at first use in the running text, and Appendix A.1 collects all of them in a single glossary table for quick reference.

2 Data

2.1 The survey

The data come from a cross-sectional household survey: each respondent was interviewed exactly once, in person, in Mandarin Chinese. Interviews were conducted in two counties bordering Giant Panda National Park, Sichuan Province – Baoxing (15–31 August 2024) and Qingchuan (10 March–10 April 2025) – seven months apart. The analysis data set supplied by the project partner contains 625 respondents and 57 columns (an identifier, three location labels, and 53 survey variables): 160 respondents from Baoxing and 465 from Qingchuan. Because the two counties were not surveyed concurrently, place of residence and survey timing cannot be statistically separated from each other in anything that follows; every model in this report treats them as a single combined *site* covariate for that reason.

The questionnaire covered five thematic blocks: (1) sociodemographics (age, gender, education, annual income, household assets, cropland area); (2) conflict experience (frequency, duration, severity, species diversity, and years of exposure to wild boar and other wildlife); (3) economic and intangible costs (estimated damaged crop area and economic loss, alongside a general attitude rating and a free-text three-word description of wild boar); (4) compensation and mitigation (application for compensation, satisfaction with the amount received, mitigation strategies used, and willingness to pay for wildlife-damage insurance); and (5) tolerance, measured with four Likert items described in Section 3.1.

2.2 Compensation status is confounded with county

The central grouping variable throughout this project is *compensation participation* (construct code *CE0*): a binary indicator of whether a household applied for compensation for wild boar damage (277 applicants, 348 non-applicants). Before any model could treat CE0 as a meaningful predictor, we checked whether it was confounded with the county/survey-wave split described above. It is, strongly: 119 of 160 respondents in Baoxing (74%) reported applying for compensation, versus only 158 of 465 in Qingchuan (34%). A Pearson chi-squared test of independence on the resulting 2×2 table (county $\times$ applied for compensation) confirms this gap is far larger than sampling noise would produce. For each cell i of the table, let O_i be the observed count and E_i the count expected under independence (row total times column total, divided by the grand total); the test statistic

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \tag{1}$$

is compared against a χ^2 distribution with $df = (r-1)(c-1) = 1$ degree of freedom for an $r = c = 2$ table ($\chi^2 = 77.1$, $p < .001$, Yates' continuity-corrected; uncorrected $\chi^2 = 78.7$; Fisher's exact test $p = 5.8 \times 10^{-19}$). The corresponding odds ratio is 5.64: a Baoxing resident had 5.6 times the odds of having applied for compensation, compared with a Qingchuan resident. Because county and survey wave move together with compensation

status, every model in Sections 4 and 5 that includes CE0 as a predictor also includes site as an explicit control.

2.3 Compensation recipients differ in assets, not in who they are

We also compared the two compensation groups on the sociodemographic and asset variables available for the full sample, using a Wilcoxon rank-sum (Mann-Whitney U) test for each continuous variable and a chi-squared test for the one categorical variable, gender. This is a covariate-balance check in the spirit of a baseline comparison table in an observational study: it asks whether recipients and non-recipients of compensation are otherwise similar households, not whether these variables predict compensation uptake in a multivariate sense. Pooled over both counties, recipients have significantly more total cropland, more total livestock, and more household assets (all $p < .001$), and a higher share of income from farming ($p = .017$), whereas age ($p = .401$), gender ($p = .318$), education ($p = .383$), and annual income ($p = .599$) show no significant difference between the two groups. Read on its own this would suggest that compensation is claimed against exposure. The next two paragraphs show that this reading is only partly supported, because these pooled tests do not account for county.

A direction-of-analysis clarification is worth making explicit here, since it is easy to read the comparison above the wrong way round. The tests just reported ask, for a variable already split into two known groups (applied for compensation vs. not), whether that variable's *distribution* differs between the groups – for example, "do cropland holdings differ between applicants and non-applicants?" They do not ask the reverse, predictive-sounding question – "if a respondent is male, is he more likely to apply for compensation?" – which would instead treat compensation status as an outcome to be predicted *from* gender, age, and the other variables. Both questions are legitimate, but they are different questions, and this balance check deliberately asks the first one: its purpose is to establish whether the two compensation groups are otherwise comparable before any later comparison between them is trusted, in the same way a baseline-characteristics table works in any observational study. It is not an attempt at a predictive model of who applies for compensation, and should not be read as one.

The second, predictive-direction question does have an answer in this report – it is simply asked separately, inside the main structural model, rather than at this bivariate, exploratory stage. The full model (Section 4.2) includes explicit paths from every sociodemographic variable *to* compensation participation (CE0), estimated jointly with every other path in the model. Table 3 reports them: household income is the one variable that significantly predicts compensation participation ($\beta = 0.080$, $p = .047$); gender is close to but does not cross the conventional threshold ($\beta = 0.069$, $p = .060$); age, education, and household assets do not reach significance. So, to answer the specific example directly: being male is associated with a somewhat higher chance of having applied for compensation, but that association is borderline, not statistically significant at the conventional 5% level, once the other background variables are accounted for simultaneously.

Checking the pooled Wilcoxon results in the predictive direction. Because the rank-sum tests above are bivariate and pooled over two very different counties, we checked them in two further ways (Table 2). First, we repeated each test within each county. The pooled differences in livestock and household assets disappear in Qingchuan ($p = .51$ and $p = .75$) and are present only in Baoxing, so they largely reflect county, not a general link between farm size and applying. Second, we fitted a logistic regression of compensation participation on these variables and gender together (Table 2). Writing π_i for the probability that household i applied, the model without county is

$$\log \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 \,\text{age}_i + \beta_2 \,\text{male}_i + \beta_3 \,\text{education}_i + \beta_4 \,\text{household size}_i + \beta_5 \,\text{income}_i + \beta_6 \,\text{agricultural share}_i + \beta_7 \,\text{cropland}_i + \beta_8 \,\text{livestock}_i + \beta_9 \,\text{assets}_i, \quad (2)$$

and the model with county adds the term $\beta_{10}\,\text{Baoxing}_i$, an indicator that is 1 for Baoxing and 0 for Qingchuan. Income, cropland, livestock, and assets enter as $\log(1 + x)$; all predictors except the two indicators are standardised, so $\exp(\beta_j)$ is the odds ratio for a one-standard-deviation increase. Without county, livestock is the clearest predictor (odds ratio 1.63 per standard deviation, $p < .001$). Once county is added, livestock is no longer significant, county itself is by far the strongest predictor (odds ratio 8.7), and cropland (odds ratio 1.66, $p < .001$) is clearly positively associated with applying. Income (odds ratio 1.26, $p = .017$) and household assets (odds ratio 0.65, $p = .010$) point in opposite directions and are only borderline once the number of tests is accounted for (Table 1); the opposite signs most likely reflect collinearity among the three wealth measures, so we do not interpret them one by one. The consistent conclusion is narrower than our first reading: county is the dominant difference between applicants and non-applicants, and, within county, larger cropland holdings go with a higher chance of having applied.

How many tests this rests on, and what we finally claim. The comparison of the two compensation groups involved several rounds of testing, and we state them all so that the reader can judge how much weight a single p-value deserves (Table 1). The first round, the balance table, consists of 10 tests: eight Wilcoxon rank-sum tests, one chi-squared test for gender, and one for county. The within-county repetition adds 16 Wilcoxon tests (eight variables in each of two counties), and the two logistic regressions add 19 coefficient tests (nine without county, ten with it), 45 tests in all. The five paths from background variables to compensation participation in the structural model are part of the main model and are not counted here. None of the tests was adjusted for multiplicity in the original balance table. With 10 tests at the 5% level, the chance of at least one false positive would be up to about 40% if no variable truly differed, so we applied two standard corrections within each round. The Holm procedure (Holm, 1979) controls the probability of making *even one* false positive among the m tests of a round: the p-values are sorted from smallest to largest, the smallest is multiplied by m, the second by $m - 1$, and so on, and a result counts as significant only if its adjusted value and all smaller ones are below .05. The Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) controls the less strict *false discovery rate*, the expected share of false positives among the results declared significant: the i-th smallest p-value is multiplied by m/i, giving an adjusted value q. Holm suits a setting where a single false claim is costly; Benjamini–Hochberg

suits a screening setting such as this one, where we look at many variables and accept that a small share of the flagged ones may be chance. We use Benjamini–Hochberg as the main criterion and report Holm where the two disagree. For example, the share of income from agriculture has the fifth-smallest of the 10 p-values in the first round ($p = .017$): its Benjamini–Hochberg value is $.017 \times 10/5 = .034$, and its Holm value is $.017 \times 6 = .10$. This does not change the main conclusions: in the first round, cropland, livestock, assets, and county remain significant, and the share of income from agriculture, which was only nominally significant ($p = .017$), remains so at a 5% false-discovery rate ($q = .034$) but not under the stricter Holm correction ($p = .10$); we therefore give it little weight. In the logistic regression with county, cropland ($q < .001$) and county ($q < .001$) are firm, while assets ($q = .033$) and income ($q = .041$) are borderline.
Our final claims about who applies for compensation are therefore these.

1. County is by far the largest difference between applicants and non-applicants (odds ratio 8.7 in the joint model; $\chi^2 = 77.1$).

2. Larger cropland holdings go with a higher chance of having applied, both within each county and in the joint model. This is the only farm-related variable for which the evidence is consistent.

3. The pooled differences in livestock and household assets are mostly a county effect: they are absent in Qingchuan and present only in Baoxing. We therefore do *not* claim that compensation tracks livestock or asset exposure in general.

4. Age, gender, and education show no reliable difference between the groups, and annual income and the share of income from agriculture none that we would interpret.

5. Compensation status is therefore not standing in for wealth or demographics in general, but we cannot separate it from county and, to a lesser extent, from the size of a household's cropland, which is why both are kept as controls in the main model.

2.4 Skewness, missing values, and outliers

Several monetary and area variables – insurance willingness-to-pay, income, household assets, and damage-related economic loss – are heavily right-skewed (household assets alone range from 77 to 3,087,729 RMB). All four are log(1+x)-transformed before entering any model, so that a handful of very large landholders do not dominate otherwise linear relationships.
Is this transformation necessary? For our conclusions, it is not. PLS-SEM makes no normality assumption (Section 4.1), so skewness is not a modelling requirement here. We therefore re-ran the observed-score cross-check regression of tolerance (Section A.6) with the money variables left raw, rank-transformed, and log-transformed. The coefficients that carry our conclusions are essentially identical in all four versions: compensation participation -0.165 to -0.168, intangible cost -0.29 to -0.30, intangible benefit $+0.11$ (Table 13). Only a borderline coefficient moves: household assets, whose p-value is .026

Table 1: Tests behind the comparison of compensation groups. Significant means significant at 5% after Benjamini–Hochberg correction within the round.

Round	**Tests**	**Significant**	**Which**
Balance table (pooled): 8 Wilcoxon, 2 chi-squared	10	5	Cropland, livestock, assets, share of agricultural income, county
Wilcoxon within Qingchuan	8	2	Education, cropland (both $q = .043$)
Wilcoxon within Baoxing	8	5	Age, income, cropland, livestock, assets
Logistic regression without county	9	1	Livestock
Logistic regression with county	10	4	County, cropland; assets and income borderline
Total	**45**		

Table 2: Who applies for compensation? Pooled and within-county Wilcoxon rank-sum p-values, and odds ratios (per standard deviation; money variables log-transformed) from a logistic regression of compensation participation on all variables jointly, including county, with heteroscedasticity-robust confidence intervals.

	Wilcoxon p			**Odds ratio**
Variable	Pooled	Qingchuan	Baoxing	[95% CI]
Age	.401	.884	.002	0.94 [0.76, 1.15]
Education	.383	.019	.091	1.22 [1.00, 1.50]
Household size	.270	.157	.552	0.82 [0.68, 0.99]
Annual income	.599	.567	.006	1.26 [1.04, 1.53]
Share of income from agriculture	.017	.274	.306	1.04 [0.85, 1.27]
Total cropland	$< .001$	.017	$< .001$	1.66 [1.28, 2.16]
Total livestock	$< .001$	.508	$< .001$	1.21 [0.96, 1.53]
Household assets	$< .001$	.746	.004	0.65 [0.47, 0.90]
Gender (male)	—	—	—	1.40 [0.93, 2.10]
County (Baoxing)	—	—	—	8.68 [4.94, 15.2]

Table 3: Sociodemographic paths to compensation participation (CE0), from the full structural model (Section 4.2), estimated jointly with every other path in the model.

Predictor	β	p
Gender	0.069	.060
Age	−0.017	.690
Education	0.069	.100
Income (log)	0.080	.047
Household assets (log)	−0.006	.900

with the log transform and .064 raw, a non-significant coefficient in the structural model in either case.

The transformation was a precaution against a few extreme households (assets up to 3 million RMB, against a typical range of a few thousand) influencing coefficients, decided from each variable's raw distribution before any model was fit and applied uniformly to all four variables. It is a harmless choice, not an important one, and the results would be the same without it. It is also imperfect: because income has zeros, $\log(1 + x)$ turns its right skew (skewness 3.0) into a left skew (-3.7).

Missingness in this dataset is not uniform, and treating all of it the same way would have been a mistake. Two compensation-experience items (timeliness and payment satisfaction, columns CE1 and CE2) are missing for the 348 respondents who never applied for compensation – by survey design (skip logic), not data loss – and are therefore excluded from the main structural model entirely, used only in supplementary applicant-only checks. Nine respondents' lowest-accepted-percentage answers and four respondents' expected-percentage answers (columns CA1, CA2) exceed 100% of assessed loss (up to 800%); these are kept as genuine responses rather than capped or deleted, since a respondent can coherently report wanting more than their assessed loss, and a winsorised (capped-at-100%) version was run separately as a sensitivity check with essentially unchanged results. Zero-valued responses that are legitimate answers – prevention effectiveness, channel count, and compensation-anticipation variables that define zero as "used nothing" or "expects nothing" – were likewise kept untouched. Apart from the two skip-logic columns, and one respondent whose village label is missing (the village label is not used in any model), the dataset has no missing values in the other variables for all 625 respondents.

Finally, we screened for multivariate outliers using Mahalanobis distance across the 15 core indicators of the Wildlife Tolerance Model (seven conflict- experience items, two cost items, two benefit items, and four tolerance items). For respondent i with indicator vector $x_i \in \mathbb{R}^{15}$, sample mean vector $\bar{x}$, and sample covariance matrix S, the squared Mahalanobis distance is

$$D_i^2 = (x_i - \bar{x})^\top S^{-1}(x_i - \bar{x}). \tag{3}$$

The matrix S^{-1} rescales and decorrelates the 15 indicators before the squared distance is taken, which is what lets D_i^2 account for how the indicators correlate with each other rather than treating each one's distance from the mean independently. Under multivariate normality, D_i^2 approximately follows a chi-squared distribution with 15 degrees of freedom (one per variable, by the same logic as Equation 1); we flagged respondents whose distance exceeded the $p < .001$ cutoff ($D^2 > 37.7$). Fourteen of 625 respondents (2.2%) were flagged – split evenly across county and compensation status (seven from each county, seven with and seven without compensation), so the flagged cases are not concentrated in any one group. These respondents were kept in the primary analysis (a statistical outlier is not automatically a data error), and every core structural result reported in Section 4 was re-estimated excluding them as a robustness check: all paths were stable in sign, magnitude, and significance.

3 Measurement

3.1 How is tolerance measured?

Tolerance was measured with four Likert items (1–5 scale), following the original Wildlife Tolerance Model questionnaire design (Kansky et al., 2016):

- *Tol1*: support for non-lethal management strategies (e.g. relocation, fencing) – positively framed.
- *Tol2*: agreement that the government should take stricter measures to control the wild boar population – reverse-coded, so that a higher raw score corresponds to lower tolerance.
- *Tol3*: perceived threat to personal safety from wild boars – reverse-coded.
- *Tol4*: willingness to accept some damage caused by wild boars – positively framed.

After reverse-coding, all four items are oriented so that 1 indicates low tolerance and 5 indicates high tolerance. Conceptually the four items are not interchangeable: they capture management preferences, policy attitudes, fear, and damage acceptance separately, which is one reason the original model treats tolerance as a multidimensional construct rather than a single attitude.

3.2 Exploratory and confirmatory checks rule out one item

Before any structural model was fit, we checked whether the four tolerance items actually behave as a single, internally consistent scale, using three independent methods of increasing formality.
First, a simple reliability check. For a k-item scale with item variances σ_i^2 and total-score variance σ_X^2 (the variance of the row-wise item sum $X = \sum_i \text{item}_i$), Cronbach's alpha is

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{i=1}^{k}\sigma_i^2}{\sigma_X^2}\right). \tag{4}$$

Equivalently, in terms of the average pairwise inter-item correlation $\bar{r}$, $\alpha = k\bar{r}/(1 + (k-1)\bar{r})$: a scale whose items correlate strongly with each other has a total-score variance much larger than the sum of the individual item variances, and α close to 1; a scale whose items are unrelated has α close to 0. The original four-item tolerance scale has low internal consistency by this measure (Cronbach's $\alpha = 0.34$, well below the conventional 0.70 threshold for a reliable scale), and Tol3's item-rest correlation (its correlation with the sum of the other three items) is -0.004 – essentially unrelated to the rest of the scale. Second, exploratory factor analysis (EFA; R package `psych` (Revelle, 2026), with rotations from `GPArotation` (Bernaards and Jennrich, 2005)) agrees: Tol3's loading on a single factor is -0.009, close to zero. Third, confirmatory factor analysis (CFA; R packages `lavaan` (Rosseel, 2012) and `semTools` (Jorgensen et al., 2026), WLSMV estimator for ordinal indicators) confirms this independently, in a formal one-factor model with a

significance test per item: Tol3's standardised loading is 0.006 with $p = .896$, statistically indistinguishable from zero, while Tol1, Tol2, and Tol4 all load significantly (Table 4). One feature of this confirmatory model must be disclosed. The standardised loading of Tol2 is 1.04, which is larger than 1 and implies a negative residual variance (-0.09) for that item: an improper solution, known as a Heywood case. It arises because the factor is defined almost entirely by the one strong correlation in the block, between Tol1 and Tol2 (polychoric $r = 0.51$), while Tol4 relates only moderately to Tol2 (0.37) and weakly to Tol1 (0.17). The individual loadings in Table 4 should therefore not be over-interpreted. The conclusion about Tol3 does not depend on it: in a proper solution, obtained by constraining the Tol1 and Tol2 loadings to be equal, Tol3 still loads at 0.03 ($p = .50$), and with a robust maximum-likelihood estimator that treats the items as continuous it loads at -0.01 ($p = .78$). The improper solution is, however, further evidence that tolerance is a weak scale, dominated by two items, and this is a limitation of every result that uses it. Three methods that make different assumptions – a bivariate correlation, an exploratory factor model, and a confirmatory factor model with a formal hypothesis test – independently agree that Tol3 does not belong with the other three tolerance items. Table 4 reports the full item-level detail behind this conclusion.

Table 4: Item-level reliability and confirmatory factor analysis (CFA) results for the four original tolerance items. Item-rest r is each item's correlation with the sum of the other three (n=625). CFA loading is the standardised factor loading from a single-factor confirmatory model on all four items (lavaan, WLSMV estimator, ordinal indicators).

Item	**Statement**	**Item-rest r**	**CFA loading**	z	p
Tol1	Supports non-lethal management	0.324	0.49	(ref.)	—
Tol2	Govt should not crack down (rev.)	0.387	1.04	4.79	$< .001$
Tol3	Perceived personal-safety threat (rev.)	-0.004	0.006	0.13	.896
Tol4	Accepts some damage	0.150	0.35	9.20	$< .001$

Figure 1 shows the same pattern graphically: Tol3's bars are flat on the floor on both measures, while the other three items are clearly elevated. On this basis we defined the primary tolerance construct as the three-item composite Tol1+Tol2+Tol4, and kept the original four-item version only as a named robustness variant. Dropping Tol3 raises reliability from $\alpha = 0.34$ to $\alpha = 0.50$ (ordinal $\alpha = 0.62$, computed from polychoric rather than Pearson item correlations, which better suit five-category ordinal data); every downstream structural path reported in Section 4 is materially unchanged between the three-item and four-item specifications. We also computed the reliability and validity statistics that a reflective PLS-SEM construct is usually expected to report: for the three-item tolerance scale, composite reliability is 0.781 and average variance extracted (AVE) is 0.551, both above their conventional 0.70 and 0.50 thresholds respectively, even though Cronbach's α (0.577 as reported by the PLS-SEM software, which standardises the items; 0.50 on the raw items) and ρ_A (0.620) remain below 0.70. This split is expected for a short, unevenly-loading three-item composite (Tol2 loads far more strongly than

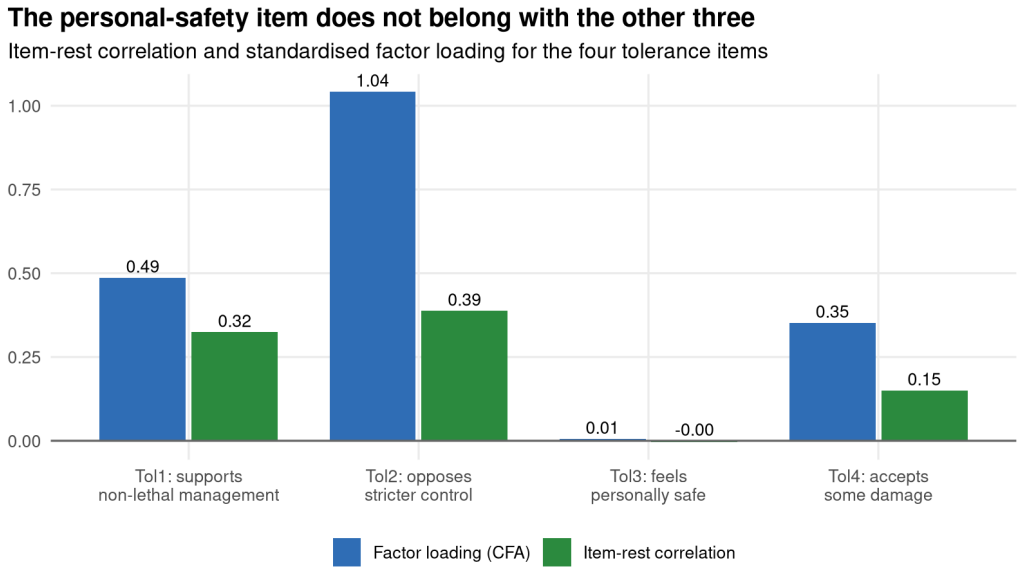


Figure 1: Item-rest correlation and standardised CFA loading by tolerance item. Tol3 is indistinguishable from zero on both measures. The Tol2 loading above 1 is the improper solution discussed in the text.

Tol4) and is a known limitation of α and ρ_A for such scales, not a contradiction of the CFA result above. Figure 2 shows the progression from the original four-item raw alpha through the two three-item variants, alongside the conventional 0.70 threshold.

3.3 Formative versus reflective: intangible cost

A second measurement question concerns how intangible cost (IC) – a respondent's general evaluative attitude toward wild boar, together with a spontaneous emotional reaction captured in a three-word description – should be modelled: as a *reflective* construct, whose indicators are interchangeable symptoms of one underlying attitude (as tolerance is treated above), or as a *formative* construct, whose indicators are distinct ingredients that combine to define the construct, and need not correlate with each other at all.
The original questionnaire design specifies intangible cost, alongside conflict experience, tangible cost, compensation anticipation, and policy support, as formative by default. We tested this assumption in two ways. First, EFA on intangible cost's own items shows weak-to-absent common-factor structure (KMO = 0.51; parallel analysis suggests essentially zero underlying factors) – the items do not behave like correlated symptoms of one hidden cause, which is expected and unproblematic for a formative composite, but would be a problem if IC were meant to be reflective. Second, and more directly, we forced intangible cost into a reflective specification and tested its discriminant validity against tolerance using the heterotrait-monotrait ratio of correlations (HTMT; Henseler et al., 2015). For this test the reflective intangible-cost factor used its two complete items, the general attitude rating and the first word rating; the second and third word ratings were left out because many respondents gave fewer than three words. The result was HTMT = 1.17, above the conventional 1.0 ceiling: under a reflective assumption, the model

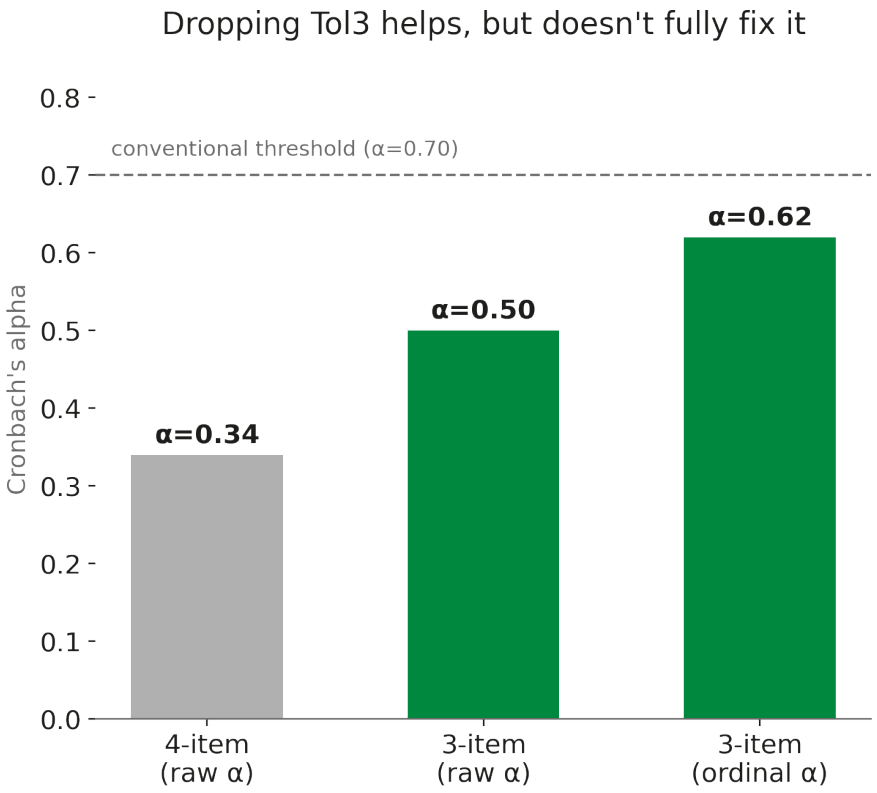


Figure 2: Tolerance scale reliability (Cronbach's α): four-item raw, three-item raw, and three-item ordinal (polychoric), against the conventional 0.70 threshold.

cannot statistically distinguish intangible cost's items from tolerance's items at all. This second, more aggressive test was reserved for intangible cost specifically – rather than for tangible cost or the other formative constructs – because IC's items are the ones conceptually closest to tolerance itself (an attitude and an emotional reaction, rather than a physical damage measure), making it the one construct where a measurement-model mistake could plausibly conflate a predictor with the outcome it predicts.
Taken together, both checks point the same way: intangible cost is retained as a formative composite, as originally specified. Its indicators are ingredients that build the concept, not interchangeable symptoms of it, and treating it as reflective would have been the wrong measurement choice – not merely a weaker one.

3.4 Why tolerance is reflective and intangible cost is formative

Treating one construct as reflective and another as formative is not an inconsistency: the choice is made construct by construct, and it is a conceptual decision about the direction of the relationship between a construct and its items, not a statistical one. The standard criteria (Jarvis et al., 2003) ask four questions. Does the construct cause the item responses (reflective), or do the items together make up the construct (formative)? Are the items interchangeable? Should they correlate with each other? And does dropping an item change what the construct means?
For tolerance, the Wildlife Tolerance Model (Kansky et al., 2016) treats the items as expressions of one underlying disposition: a more tolerant resident is expected to support non-lethal management, oppose stricter control, and accept some damage, all at once. That is the reflective logic, and it is why we could test it with factor analysis and drop Tol3 without redefining the construct. For intangible cost, the items are a general attitude rating and the valence of up to three freely chosen words. The original model specifies this block, like the other cost and experience blocks, as an index built from its components, which need not correlate.
We do not want to overstate how clean this distinction is. The empirical checks are compatible with our choice but do not prove it. Tolerance's reliability is modest even after dropping Tol3 (Section 3.2), and its three items cover different content, so a formative reading of tolerance is defensible too. Conversely, an attitude rating and word valences could reasonably be read as expressions of one attitude, that is, as reflective. Weak correlations among items do not by themselves make a construct formative. For this reason we checked whether the choice matters. Re-estimating the full model with tolerance weighted formatively (PLS mode B) instead of reflectively (mode A) leaves the results unchanged: the compensation path is -0.108 instead of -0.109, intangible cost -0.334 instead of -0.338, intangible benefit $+0.162$ instead of $+0.156$, prevention effectiveness $+0.107$ instead of $+0.113$, all still significant, and R^2 is 0.235 instead of 0.234. The alternative for intangible cost was tested in Section 3.3: as a reflective construct it cannot be separated from tolerance. Our conclusions therefore do not depend on the reflective/formative labels, but the closeness of intangible cost to tolerance is a limitation in either specification (Section 6.3).

3.5 Feature engineering

Three engineered features feed into the main model. First, as described above, tolerance enters as the three-item composite Tol1+Tol2+Tol4. Second, all skewed monetary and area variables (household income, household assets, economic loss from damage, insurance willingness-to-pay, damaged crop area) are log(1+x)-transformed. Third, survey site (Baoxing vs. Qingchuan) is engineered as an explicit binary covariate, because Section 2.2 showed that compensation participation is strongly confounded with county; without this covariate, county differences would be invisible to the model and could be misattributed to compensation status or conflict experience.

We additionally engineered, but did not use as a control in the primary model, a household wealth/livelihood index via principal component analysis (PCA; R package `factoextra` (Kassambara and Mundt, 2026)) on 15 previously-unused variables (crop areas by type, livestock counts by type, and total land/livestock/asset value). The first principal component explains 27.9% of the variance in these 15 variables and correlates strongly with total household assets ($r = 0.75$) but only weakly with annual cash income ($r = 0.17$), indicating that income is a poor proxy for asset wealth in this sample. This index does not predict tolerance directly ($p = .33$), but it does predict self-reported economic loss strongly (unstandardised coefficient 0.28 on log loss, $p < .001$): wealthier households have more agricultural assets at risk and report more loss accordingly. The original Wildlife Tolerance Model used a single manifest item (crop asset value) as its only wealth control; this index shows that a richer, PCA-based wealth measure is available, though it functions as a socioeconomic control rather than a new psychological construct, and was therefore not added to the primary structural model reported in Section 4.

4 Main Model and Findings

4.1 Why partial least squares structural equation modelling

We use partial least squares structural equation modelling (PLS-SEM, R package `seminr` (Ray et al., 2026)) to estimate the full structural model in one step. PLS-SEM was chosen for three reasons specific to this data. First, it accommodates formative and reflective constructs simultaneously within a single estimation: tolerance is reflective (Section 3.1), while conflict experience, tangible cost, intangible cost, compensation anticipation, and policy support are formative (Section 3.3). Second, it does not require multivariate normality, which is convenient given the skewed monetary variables described in Section 2.4. Third, it remains reliable at our sample size ($n = 625$) with a comparatively complex construct structure.

4.2 Model specification

The structural model links four layers of constructs: a background layer (conflict experience with wild boar and with all species, survey site, national-park residence, and sociodemographics – gender, age, education, income, household assets); an exposure layer (prevention effectiveness and compensation participation, CE0); a mediator layer (compensation anticipation, policy support, tangible cost, intangible cost, and intangible benefit); and the outcome, tolerance. Every construct in an earlier layer has a direct structural path to every construct in a later layer, including a direct path from each background variable to tolerance itself; this dense specification is what lets us report, for example, that a given predictor's effect on tolerance is estimated *after* already accounting for site, national-park residence, and demographics in the same equation, rather than as a separate, unadjusted comparison. Skewed monetary indicators enter log-transformed within their formative blocks, exactly as in Section 3.5. All paths are estimated jointly and bootstrapped with 5,000 resamples to obtain standard errors and p-values for every path simultaneously. Because the underlying data are cross-sectional and observational, every coefficient below should be read as an *association*, adjusted for the other variables in the same model, not as a causal effect.

4.3 RQ1: does compensation predict tolerance, and is the effect mediated?

The full model explains a meaningful share of the variance in tolerance ($R^2 = 0.234$), driven predominantly by intangible cost and, to a lesser extent, intangible benefit and prevention effectiveness. Table 5 reports the four direct predictors with a significant path to tolerance.

Figure 3 shows all four paths together; Figure 4 shows every direct path to tolerance in the model, including those that are not significant. Intangible cost has the strongest association with tolerance of any predictor in the model ($\beta = -0.34$, $p < .001$): residents who perceive more intangible cost from wild boar report markedly lower tolerance. Intangible benefit ($\beta = +0.16$, $p < .001$) and prevention effectiveness ($\beta = +0.11$, $p = .005$) both associate positively with tolerance. Compensation participation itself has a smaller, but

Table 5: Direct predictors of tolerance with a significant path in the full structural model ($R^2 = 0.234$; $n = 625$; 5,000 bootstrap resamples).

Predictor	β	p
Intangible cost (IC)	-0.34	$< .001$
Intangible benefit (IB)	$+0.16$	$< .001$
Prevention effectiveness (PE)	$+0.11$	.005
Compensation participation (CE0)	-0.109	.003

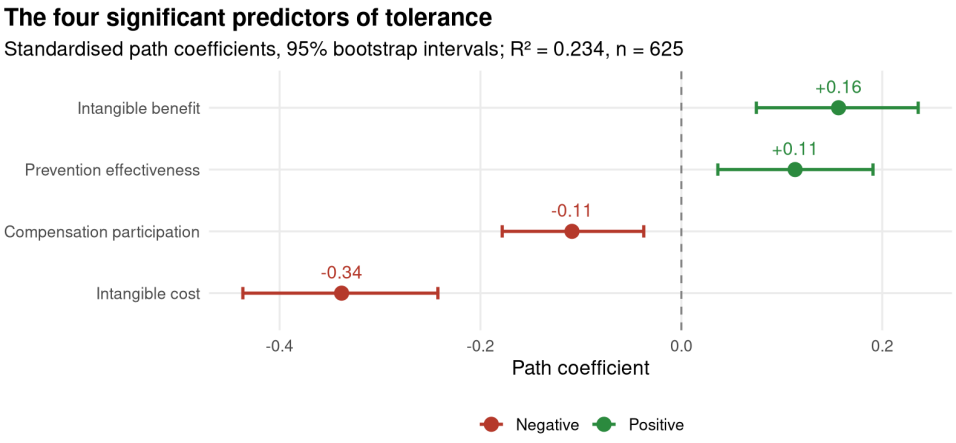


Figure 3: Standardized path coefficients from the four significant direct predictors of tolerance, with 95% bootstrap intervals. Red: negative association; green: positive association.

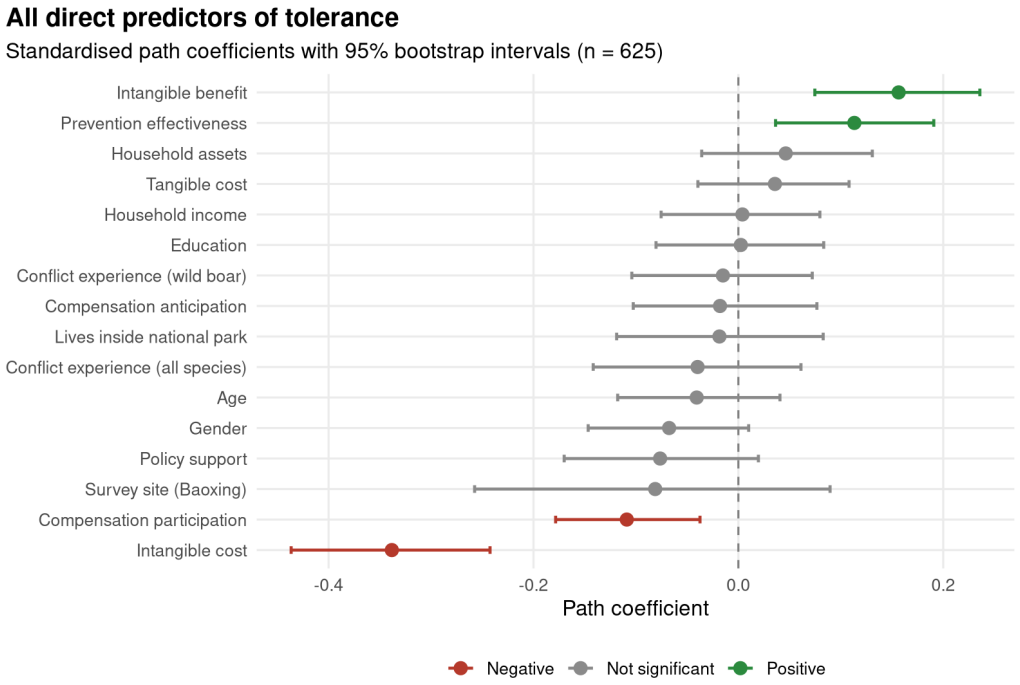


Figure 4: All direct paths to tolerance in the full structural model, with 95% bootstrap intervals. Green: positive; red: negative; grey: interval includes zero.

still significant, negative association with tolerance ($\beta = -0.109$, $p = .003$), even after adjusting for survey site – we refer to this as the *compensation paradox* throughout this report, since it runs opposite to the policy assumption that compensation should raise tolerance.

A conceptual point is worth stating clearly before going further, because the next result can otherwise look like it contradicts the one just given. Section 4.3 above already established that compensation participation has a real, statistically robust association with tolerance (Table 5; and, as Section 5 shows, that association is negative in all 8,192 alternative model specifications we tested). That is one question – *does* compensation participation associate with tolerance? – and it has a clear answer: yes. Mediation analysis asks a second, logically separate question: *through which of our measured variables* does that association travel? An effect can be real, large, and completely robust while none of a specific, finite list of candidate mediators explains it – this happens whenever the true mechanism runs through something the study did not measure. Finding no significant mediator among the five we tested therefore does not weaken or cast doubt on the direct effect established above; it means only that this particular analysis cannot yet say *why* the effect exists, which is precisely the open question this subsection reports.

With that distinction in mind: we tested whether the compensation–tolerance association runs through the five measured mediator constructs (compensation anticipation, policy support, tangible cost, intangible cost, and intangible benefit), using bootstrapped specific-indirect-effect tests (Nitzl et al., 2016). Table 6 reports the result: none of the five specific indirect paths is individually significant (all $p > .16$), and the total indirect effect summed across all five is small (-0.024), corresponding to a variance-accounted-for

(VAF) of 18% of the total compensation–tolerance association.

Table 6: Bootstrapped specific-indirect-effect tests for compensation participation → mediator → tolerance (Nitzl et al., 2016), with 95% percentile bootstrap confidence intervals. Direct effect: −0.109 ($p = .003$). Total indirect effect: −0.024 (VAF = 18%).

Code	Construct	Indirect effect	SE	95% CI	p
CA	Compensation anticipation	−0.0017	0.0046	[−0.010, 0.009]	.800
PS	Policy support	−0.0013	0.0054	[−0.013, 0.010]	.808
TC	Tangible cost	+0.0027	0.0036	[−0.003, 0.011]	.382
IC	Intangible cost	−0.0178	0.0141	[−0.048, 0.008]	.165
IB	Intangible benefit	−0.0060	0.0070	[−0.021, 0.007]	.354

In other words, most of the compensation paradox is not explained by the mediators we measured. The confidence intervals show how strong this statement is. A non-significant indirect effect is an absence of evidence, not evidence of absence: the interval for intangible cost, the most plausible mediator, runs from −0.048 to +0.008, so an indirect path of up to roughly 40% of the direct effect remains compatible with the data. With a bootstrap standard error of 0.014, this design would detect an indirect effect through intangible cost with 80% power only if it were about 0.04 or larger. The other four intervals are much tighter around zero. So this does not mean there is no mechanism connecting compensation to tolerance. Two explanations remain open. The mechanism may run through something outside this measurement model: unmeasured grievance related to the compensation process, prior conflict history not captured by our conflict- experience items, or some other shared cause behind both compensation-seeking and lower tolerance. Alternatively, the mediators may be measured too noisily to carry the effect they really transmit; we examine this second possibility in Section 6.2. This is an open question for future data collection (Section 6), not a settled mechanism.

Why do the cost constructs, which the Wildlife Tolerance Model places between compensation and tolerance, not act as mediators here? An indirect effect is the product of two legs: compensation must predict the mediator, and the mediator must predict tolerance. Figure 5 shows both legs for each construct. Intangible cost and intangible benefit clearly predict tolerance (second leg), but compensation participation does not predict either of them once county, conflict experience, and demographics are adjusted for (first leg: +0.05 and −0.04, both intervals include zero). Tangible cost fails on both legs: applicants report somewhat more damage (+0.07, not significant), and damage is unrelated to tolerance (+0.04). In plain terms, applicants and non-applicants perceive similar costs and benefits of wild boar, so those perceptions cannot be what makes applicants less tolerant. The same logic explains the absence of moderation in Section 4.5: costs and benefits relate to tolerance in the same way in both groups.

4.4 Model checks on individual paths

Three additional checks qualify how confidently individual paths above should be read. First, as already noted in Section 3.1, tolerance's reliability, while improved by dropping Tol3, still falls short of the conventional threshold by the α and ρ_A criteria; treat tol-

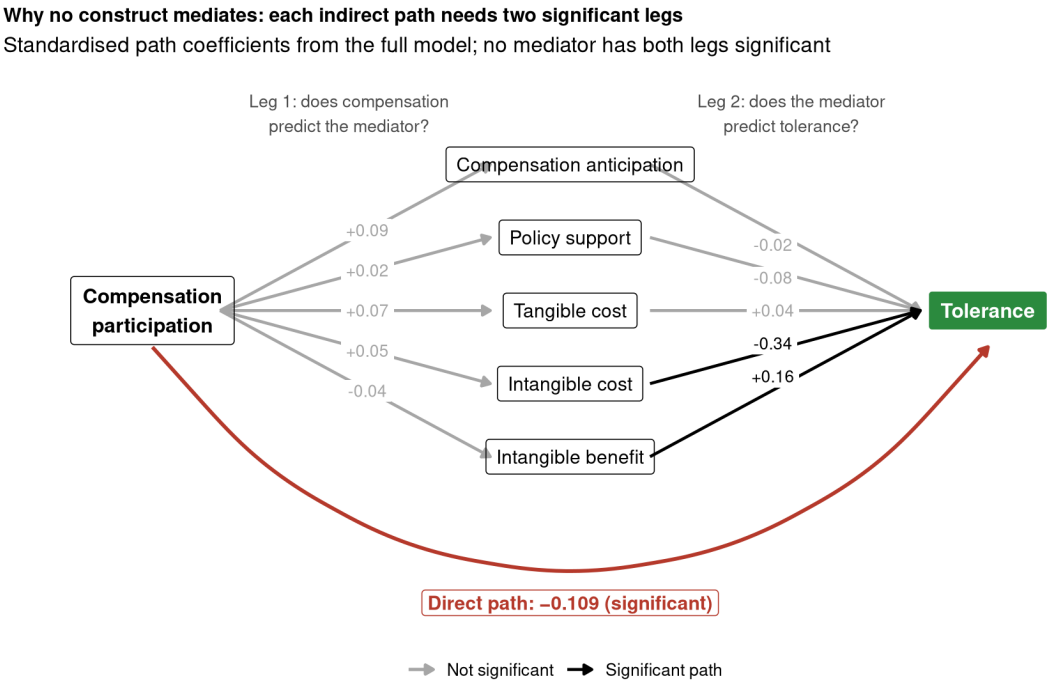


Figure 5: The two legs of each indirect path from compensation participation to tolerance. Black arrows: 95% bootstrap interval excludes zero; grey: includes zero. No construct has two black legs.

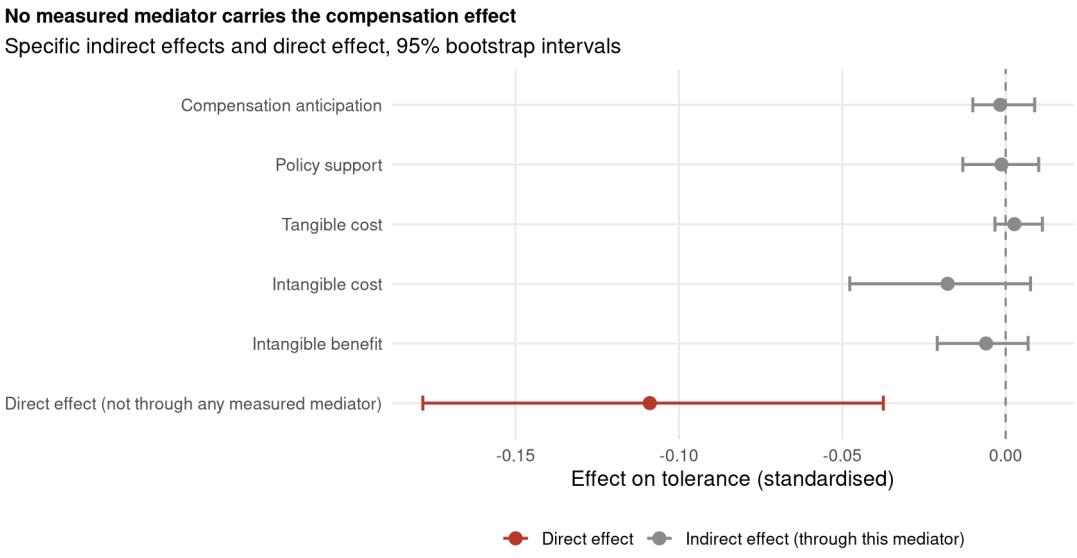


Figure 6: Specific indirect effects of compensation participation on tolerance (grey) and the direct effect (red), with 95% bootstrap intervals. All five indirect intervals include zero; the direct effect does not.

erance throughout as a meaningful but noisy measure. Second, policy support overlaps empirically with conflict experience and with survey site: HTMT is 1.01 against conflict experience and 1.22 against site, both above the 1.0 ceiling that normally signals a discriminant-validity problem. The two conflict-experience constructs also overlap with each other (HTMT 1.17). HTMT was designed for reflective constructs, and here it is applied to formative composites and single items, so these values are a warning sign rather than a formal test. We therefore do not build a strong substantive claim on the policy-support path in isolation – whatever that path's coefficient captures may be entangled with conflict experience and site rather than policy support specifically. Third, survey site significantly predicts all five mediator constructs (policy support, tangible cost, intangible cost, and intangible benefit at $p < .001$; compensation anticipation at $p = .044$), but has no significant remaining direct path to tolerance once those mediators are in the model ($\beta = -0.08$, $p = .34$). We read this as reassuring rather than concerning: site's entire detectable influence on tolerance appears to run through the mediators already in the model, rather than through some separate, unmodelled channel.

4.5 RQ2: do costs and benefits work differently after compensation?

RQ2 asks whether the relationships between costs/benefits and tolerance change depending on compensation experience – that is, whether compensation *moderates* these relationships. We tested this directly with three interaction terms (compensation participation $\times$ tangible cost, $\times$ intangible cost, and $\times$ intangible benefit) in a two-stage moderation model, rather than by comparing separately-fitted coefficients in each group. The interaction-term approach is the stronger test: comparing two separately-estimated group coefficients only tells you the two numbers look different, not whether that difference is statistically distinguishable from sampling noise in two smaller subsamples; a formal interaction term tests the difference in slopes directly, using the full sample in one joint estimation.

All three interactions are non-significant (tangible cost $\times$ compensation, $p = .216$; intangible cost $\times$ compensation, $p = .234$; intangible benefit $\times$ compensation, $p = .899$). This first moderation model contains only the three cost and benefit constructs, compensation participation, and their interactions. Because compensation participation is confounded with county (Section 2.2), we repeated the test with the full control set of the main model in the tolerance equation (survey site, national-park residence, both conflict-experience constructs, demographics, prevention effectiveness, compensation anticipation, and policy support). The conclusion is the same: $p = .312$, .142, and .871 respectively, with compensation participation itself still negative ($\beta = -0.106$, $p = .004$). An ordinary regression on observed scores with the same controls agrees, both for the single interactions (all $p > .31$) and for a joint test of all three ($F = 0.49$, $p = .69$). The cost/benefit relationships with tolerance are the same strength regardless of compensation status: intangible cost remains the strongest negative predictor and intangible benefit remains positive in both groups, and compensation experience does not alter either slope. Figure 7 plots the three interaction coefficients with their intervals, for the model without and with controls: all include zero. One exploratory observation qualifies this. If interactions of the

costs and benefits with *county* are added to the observed-score regression as well, the intangible cost × compensation term (+0.21, $p = .028$) and the intangible cost × county term (−0.26, $p = .029$) both become nominally significant, with opposite signs. These are two of six additional tests, they are not corrected for multiplicity, and compensation and county are too entangled to separate cleanly in this way, so we report it only as an indication that the strength of the intangible cost–tolerance relationship may vary across settings, not as a finding. Figure 8 shows the same question informally, as fitted lines of tolerance against each predictor separately for applicants and non-applicants. These raw lines are unadjusted, so differences in their slopes (for instance for intangible benefit) can reflect other differences between the groups; the adjusted interaction tests in Figure 7 are the formal answer.

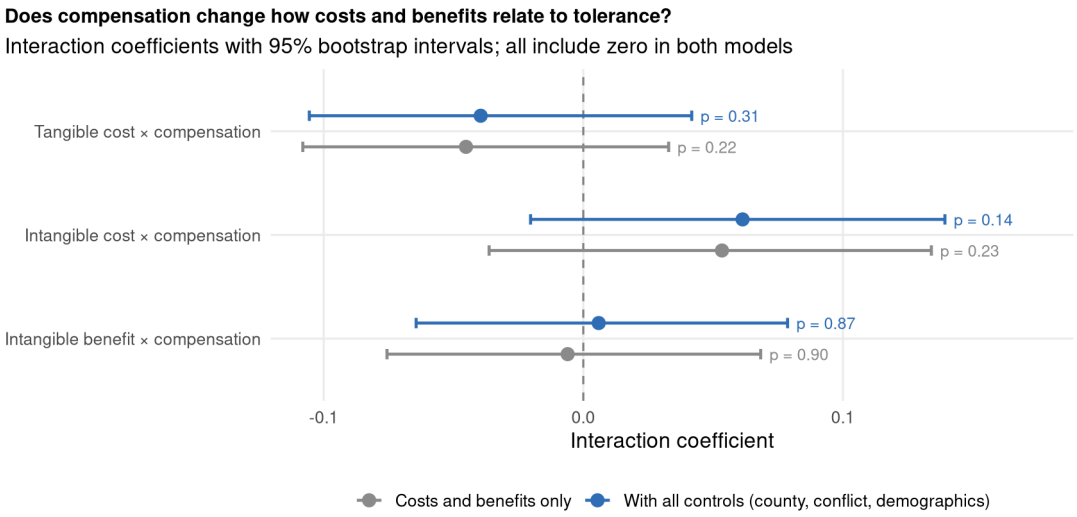


Figure 7: Interaction coefficients (compensation participation × cost or benefit) on tolerance, with 95% bootstrap intervals, from the model with costs and benefits only (grey) and the model with the full control set (blue).

4.6 Summary of the main-analysis findings

For RQ1, compensation participation is negatively associated with tolerance, and no specific measured mediator explains why. For RQ2, intangible cost is the strongest negative predictor of tolerance and intangible benefit is a positive one, and compensation status does not change the strength of either relationship. Both findings are associations from a single cross-sectional survey with an incomplete measurement model; they motivate specific follow-up questions (Section 6) rather than closing the topic.

Table 7 counts what was and was not significant. The original survey sheet has 57 columns. After removing the identifier, the township and village labels, the four tolerance items (the outcome), and the two applicant-only follow-up items, 48 variables remain as candidate covariates (county enters as survey site). Screened one at a time (Appendix A.4), 15 are associated with tolerance at $p < .05$ and 10 after controlling the false discovery rate at 5%; with county as an additional covariate, 12 survive. These counts overstate the number of

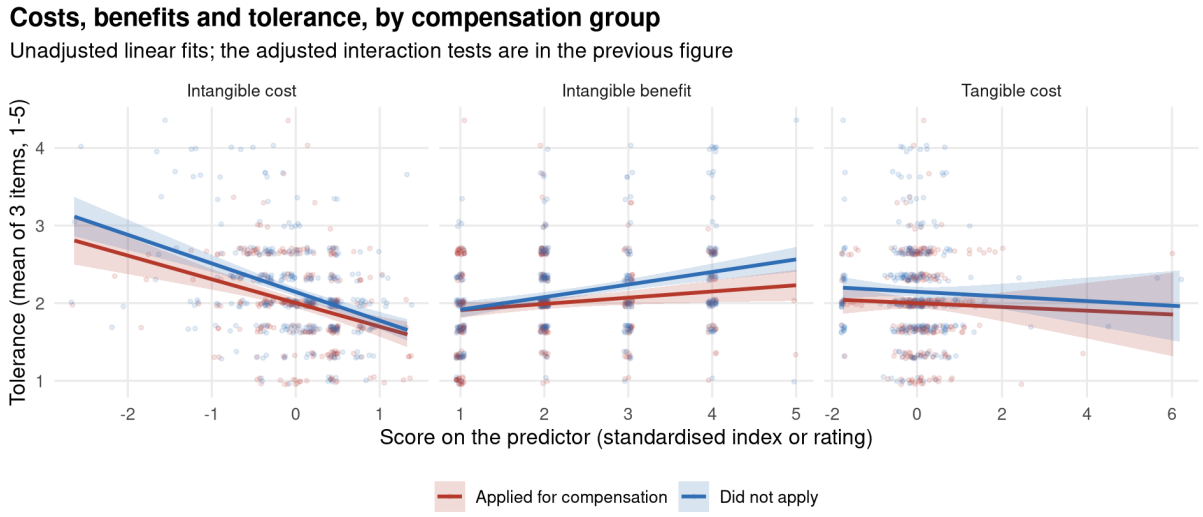


Figure 8: Tolerance against intangible cost, intangible benefit, and tangible cost, separately for households that did and did not apply for compensation (unadjusted linear fits with 95% confidence bands).

independent findings, since five of the significant variables are different indicators of the same construct (intangible cost) and several others measure conflict with other species. In the full structural model, where the variables are grouped into constructs and adjusted for each other, four constructs are significantly associated with tolerance.

Table 7: Summary of significant findings.

Question	**Count**	**Which**
Survey variables associated with tolerance, one at a time (of 48)	15 nominal; 10 after FDR control	General attitude to wild boar and word ratings (intangible cost), ecological value (intangible benefit), prevention effectiveness, lowest accepted compensation, severity and duration of conflict with all species, potato area
Constructs with a significant direct path to tolerance (of 16)	4	Intangible cost (-0.34), intangible benefit ($+0.16$), prevention effectiveness ($+0.11$), compensation participation (-0.109)
Significant mediators of the compensation effect (of 5)	0	None; total indirect effect -0.024 (18%)
Significant interactions with compensation (of 3)	0	None, without or with the full control set
Significant moderators	0 in the interaction tests	Compensation status does not moderate any cost or benefit path; the multi-group analysis finds one exploratory difference (Section 5.2)

5 Robustness and Multi-Group Analysis

RQ3 asks how robust the RQ1/RQ2 findings are to reasonable alternative analytical choices, and whether they hold across compensation groups. We address this with two independent exercises: a specification-vibration analysis across many plausible model variants, and a permutation-based multi-group analysis (MGA) comparing compensated and uncompensated residents directly. As the StaBLab consulting-methodology notes that guided this project put it, the goal of a robustness exercise is not to search for significant results, but to examine how sensitive the main findings are to different reasonable analytical choices – trying many specifications and reporting only the one that happens to be significant would be a form of selective reporting, not robustness testing.

5.1 Specification vibration

We first compared six theoretically motivated model specifications, each changing exactly one analytical choice at a time (e.g. three-item vs. four-item tolerance, alternative cost definitions, alternative control sets). We then went further and estimated every possible combination of 13 binary adjustment choices – $2^{13} = 8{,}192$ models in total – following the vibration-of-effects framework of Patel et al. (2015), who found that observational associations can flip sign (a "Janus effect") under different, individually defensible specifications; in their reanalysis of 417 variables, 31% showed such a sign flip. Across all 8,192 model variants estimated here, the compensation participation – tolerance association was negative in every single model (0 Janus effects), and reached conventional significance ($p < .05$) in 99.4% of them. The median standardized coefficient across all 8,192 models was $\beta = -0.09$, consistent with the range obtained across the six theoretically motivated specifications ($\beta = -0.128$ to -0.143) and with the primary model reported in Section 4.3 ($\beta = -0.109$). We additionally required every specification to meet minimum measurement-quality bars (formative outer loadings ≥ 0.70 where applicable, variance inflation factor < 3) and computed the standardized root mean square residual (SRMR) across all specifications to confirm that model fit did not degrade as adjustments were added. On this evidence, the negative compensation–tolerance association is not an artefact of any single analytical choice. Appendix A.5 shows an independent re-computation of the same exercise on observed (mean-score) variables.

5.2 Multi-group analysis

The second robustness exercise directly compares the structural model between the 277 residents with compensation experience (CE0 = 1) and the 348 without (CE0 = 0), using permutation-based multi-group analysis (1,000 permutations). A permutation test makes no distributional assumption and is appropriate for the two groups' unequal sizes; it tests whether path coefficients are statistically different across groups, going beyond an earlier, purely visual sub-group comparison. Before comparing any path coefficients across groups, we first confirmed that discriminant validity (HTMT) holds *within* each group separately: an HTMT violation within a subgroup would make that subgroup's path coefficients unreliable, and any between-group comparison built on them meaningless. Discriminant validity held in both groups.

We ran two versions of the comparison: MGA 1, the full model with all background controls; and MGA 2, with sociodemographic controls removed, as a confounder check – if a path difference in MGA 1 disappeared once demographics were added back and a *different* path emerged once they were removed, that would suggest the demographics were themselves confounding the group comparison. In fact, the significant findings in MGA 1 and MGA 2 do not overlap, and no previously-significant path becomes non-significant when demographics are added or removed; the confounder check passes.

The core finding, robust across both versions, is one of *structural invariance*: the model's path coefficients do not differ significantly between compensated and uncompensated residents for almost every path, including the intangible-cost – tolerance path, which is significant in both groups ($p < .001$ in each). One path is the exception: in MGA 2 (without demographic controls), conflict experience with all species has a significantly different relationship with tolerance across the two groups ($\beta = -0.187$ for compensated residents versus $\beta = +0.038$ for uncompensated residents; $\Delta\beta = -0.225$, $p = .039$). We interpret this as suggestive evidence that compensation experience may amplify the conflict–tolerance link: among compensated residents, more conflict experience associates with markedly lower tolerance, whereas among uncompensated residents the association is close to zero. We supplemented the permutation p-value, which only tells us whether a difference is statistically detectable, with Cohen's f^2 effect size, which asks how functionally important each predictor is *within* each group separately (thresholds: negligible < 0.02, small ≥ 0.02, medium ≥ 0.15, large ≥ 0.35; Cohen, 1988). In MGA 2, intangible cost's effect size is medium-to-large among uncompensated residents ($f^2 = 0.258$) and small-to-medium among compensated residents ($f^2 = 0.103$) – the same pattern of intangible cost mattering more for uncompensated residents holds across both MGA versions and both ways of quantifying "matters" (statistical difference and effect size).

Finally, the two groups differ in how much of tolerance the model explains overall: $R^2 = 0.423$ for uncompensated residents versus $R^2 = 0.299$ for compensated residents in MGA 1. The model captures uncompensated residents' tolerance more completely than it captures compensated residents' tolerance. Read alongside the mediation result in Section 4.3 – where the measured mediators account for only 18% of the compensation–tolerance association – this lower R^2 points the same direction: compensated residents' tolerance appears to be shaped by additional factors this model does not measure, plausibly related to their compensation experience itself (for example, perceptions of payment adequacy, timeliness, or fairness, which our questionnaire only captured for a subset of applicants and could not include in the main structural model; Section 2.4) or to prior conflict history not captured by our conflict-experience items. This is precisely the kind of open question a longitudinal or better-instrumented follow-up study, discussed in Section 6, would be positioned to resolve.

To summarize RQ3: the model is structurally stable across compensation groups on nearly every path, and the negative compensation–tolerance association from RQ1 does not depend on any single analytical choice among the 8,192 tested. The one genuine group difference we find – conflict experience mattering more for compensated residents – and the lower explanatory power of the model for that same group, both point toward unmeasured factors specific to the compensation experience as the most promising direction for future work.

6 Discussion

6.1 What our design can and cannot establish

Two methodological points are worth stating explicitly, because they define the boundary of what this analysis can claim.

First, several exhibits in this report – most notably the covariate-balance comparison in Section 2.3 – compare the distribution of a variable (assets, gender, and so on) *conditional on* compensation group, rather than modelling compensation status as an outcome to be predicted from those variables. This is a deliberate choice, not an oversight: its purpose is to check whether the two compensation groups are comparable before any later comparison is trusted, exactly as a baseline characteristics table does in an observational or quasi-experimental study. The complementary, predictive-direction question – do these same variables predict who applies for compensation? – is answered separately, inside the main structural model itself (Section 4.2), where sociodemographic and background constructs have explicit paths *to* compensation participation. There, only household income reaches significance ($\beta = 0.080$, $p = .047$); gender is borderline ($\beta = 0.069$, $p = .060$); age, education, and household assets are not significant predictors of compensation participation once the other background variables are accounted for jointly. Both directions of analysis appear somewhere in this report; neither substitutes for the other.

Second, every coefficient in this report is an *association*, adjusted for the other variables in the same model, not a causal effect. The data are observational and cross-sectional: compensation status was not randomly assigned, and we do not observe tolerance before a household applied for compensation. Where we write that compensation participation is "negatively associated with" tolerance, we mean precisely that and no more.

6.2 Sensitivity checks on the main finding

We ran three further checks aimed at the most likely ways in which the compensation paradox could be an artefact. All three are exploratory and were added after the main analysis; they use ordinary regression on the construct scores from the main model, or a separate latent-variable model, and are reported here to qualify the main result, not to replace it.

Does the association hold within each county? Compensation status is strongly tied to county (Section 2.2), and with only two counties a site covariate cannot fully separate the two. We therefore re-estimated the compensation–tolerance association separately in each county (Table 8). The association is negative in both. In Qingchuan it is clearly different from zero. In Baoxing the point estimate is, if anything, larger, but the confidence interval is wide and includes zero: only 41 of the 160 Baoxing respondents did not apply for compensation, so that comparison is imprecise. A formal compensation × county interaction is not significant ($p = .34$). The paradox is therefore not produced by pooling two different counties, but the statistical evidence for it rests mainly on Qingchuan.

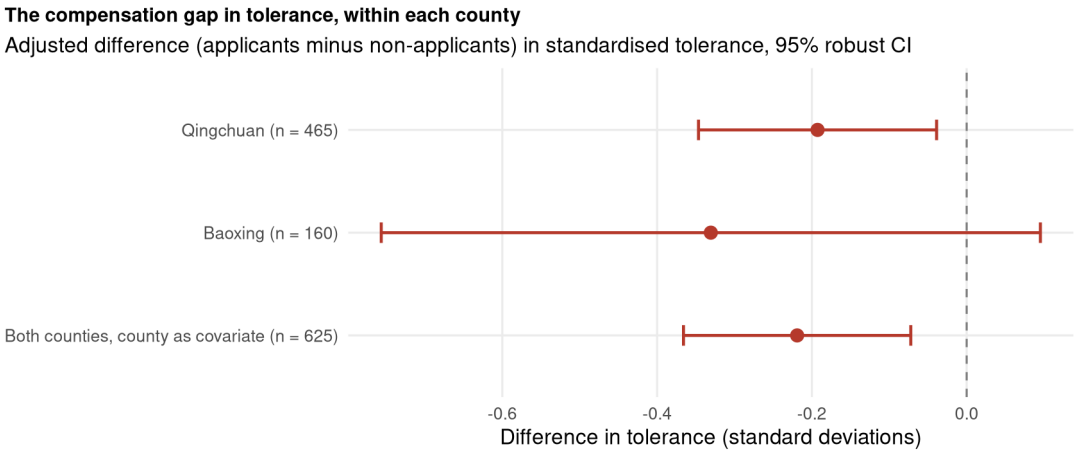


Figure 9: The adjusted compensation gap in tolerance, in each county and with both counties pooled.

Table 8: Compensation participation and tolerance within each county. Entries are the adjusted difference in the standardised tolerance score between applicants and non-applicants (in standard deviations), from a linear regression on the construct scores of the main model, controlling for the background variables and the mediator constructs, with heteroscedasticity-robust (HC3) standard errors. Because compensation participation is entered as a 0/1 variable here, these values are on a different scale from the standardised β in Table 5.

County	n	**Difference**	**95% CI**	p
Qingchuan	465	−0.19	[−0.35, −0.04]	.014
Baoxing	160	−0.33	[−0.76, 0.10]	.127
Pooled, with county as covariate	625	−0.22	[−0.37, −0.07]	.004

Could measurement error hide a mediator? Our mediators are short self-report blocks. Random measurement error in a mediator has two known consequences, which is why it is a natural suspect for "missing" mediation. First, it causes *attenuation bias*: the estimated effect of the mediator on tolerance (the second leg) is pulled towards zero, and with it the indirect effect, which is the product of the two legs. Second, it causes *incomplete adjustment*: because compensation participation is correlated with the mediator, part of the mediator's true influence is not absorbed by the noisy measure and shows up instead as a direct effect of compensation. Together these can make a truly mediated effect look direct. PLS-SEM composites do not correct for either problem.

As a sensitivity analysis we refitted the model as a latent-variable model (R package `lavaan`, WLSMV estimator), with tolerance as an ordinal latent factor and *all five* mediators entered together, each as a latent variable measured by its construct score with an assumed reliability between 1.0 (no error) and 0.6 (Table 9). The same reliability was applied to every mediator because we have no external estimate for each one. For the formative constructs, whose indicators are not assumed to be error-free copies of one quantity, this is a sensitivity assumption, not an estimated property, and the results should be read in that light.

The result is the same at every assumed reliability. Correcting for error strengthens the second leg for intangible cost considerably (standardised path -0.41 without correction, -0.65 at reliability 0.6) and the indirect effect through intangible cost grows from -0.021 to -0.044, but its interval always includes zero. The indirect effects through the other four mediators stay close to zero, as does the total indirect effect (-0.032 to -0.049, intervals including zero throughout), and the direct association between compensation and tolerance stays negative and different from zero (-0.146 to -0.129). The limiting factor is the first leg. Random error in a mediator that serves as an *outcome* of compensation does not bias that first-leg coefficient; it only widens its interval. Compensation participation is barely related to intangible cost ($+0.05$ to $+0.07$, $p \approx .22$) or intangible benefit (-0.04 to -0.05, $p = .35$), and no correction for random error can create a relationship that is not there. The two mediators with a nominally detectable first leg, compensation anticipation ($+0.09$ to $+0.12$, $p \approx .04$) and, borderline, tangible cost, have a second leg indistinguishable from zero (-0.03 and $+0.02$ to $+0.14$). We conclude that attenuation from random measurement error does not explain the absence of mediation in this data.

Two other kinds of error are not covered by this analysis and could work in the opposite direction: error that is shared between intangible cost and tolerance because both are reported by the same person (which would inflate their second leg, see Section 6.3), and error that depends on compensation status itself (differential error), for instance if applicants rate costs differently because of their dealings with the compensation process.

What the robustness analysis does and does not show. That the compensation coefficient is negative in all 8,192 specifications (Section 5) shows that the result does not depend on our analytic choices. It does not show that the association is causal, or free of bias: every specification uses the same respondents, the same self-reported measures, and the same non-random assignment to compensation, so any bias from those sources is present in all of them.

Table 9: Latent-variable sensitivity analysis for all five mediators: standardised indirect effects of compensation participation on tolerance (95% CI in brackets) under different assumed reliabilities of the mediator scores. CA: compensation anticipation; PS: policy support; TC: tangible cost; IC: intangible cost; IB: intangible benefit. Direct is the compensation → tolerance path. Values are comparable across rows, but not in size with the PLS-SEM coefficients in Table 5.

Rel.	Direct	via CA	via PS	via TC	via IC	via IB	Total indirect
1.0	−0.15	−0.003	−0.002	+0.001	−0.021	−0.008	−0.032 [−0.075, 0.010]
0.9	−0.14	−0.003	−0.002	+0.002	−0.024	−0.009	−0.035 [−0.083, 0.012]
0.8	−0.14	−0.003	−0.002	+0.004	−0.028	−0.009	−0.039 [−0.094, 0.015]
0.7	−0.13	−0.004	−0.002	+0.007	−0.034	−0.010	−0.044 [−0.108, 0.021]
0.6	−0.13	−0.005	−0.002	+0.013	−0.044	−0.011	−0.049 [−0.129, 0.031]

The 95% interval of every individual indirect effect includes zero at all reliabilities; the direct path is significant ($p < .03$) at all reliabilities.

6.3 Limitations

Five limitations bound every conclusion in this report. First, the comparison is cross-sectional: we lack each respondent's tolerance before they applied for compensation, and payment timelines are not consistently established across respondents, so we cannot rule out that some of the association we attribute to compensation participation reflects pre-existing differences between residents who chose to apply and those who did not. Second, county and survey period coincide completely (Section 2.2), and the sampling frame and any exclusions applied before we received the data are incompletely documented; we have controlled for the county/survey-wave confound statistically, but cannot rule out other, unrecorded differences between the two fieldwork rounds. Third, several measurement blocks are short and conceptually overlapping (Section 3.3), and the free-text word responses used for part of the intangible-cost measure have structural missingness; together these limit how strongly we can claim to have isolated distinct latent psychological constructs, as opposed to correlated but imperfectly separated ones. Fourth, tolerance and intangible cost are both attitudes reported by the same person in the same interview. A respondent who is generally aggrieved is likely to rate wild boar more negatively and to report lower tolerance at once, so part of the strong intangible cost–tolerance association may reflect shared reporting tendencies rather than one attitude driving the other; the discriminant-validity result in Section 3.3 points the same way. The finding that intangible cost is the strongest correlate of tolerance should be read with this in mind. Fifth, tolerance itself is measured with limited reliability (Section 3.2), and the mediation, moderation, and multi-group analyses involve many tests on one data set. We treat the compensation–tolerance path as the one confirmatory result; the mediation and moderation results are mostly null findings, for which the sample gives limited power (Section 4.3), and the single multi-group difference we report should be regarded as exploratory until replicated.

6.4 Recommendations for the next study

Three concrete steps would materially strengthen the next round of analysis or data collection. First, several fieldwork details should be clarified directly with the project partner before any further modelling: the exact routing between applying for and receiving compensation, which sampling exclusions were applied and why, precise fieldwork dates, the units underlying the area variables, and the small number of compensation- percentage responses that exceed 100% of assessed loss (Section 2.4). Second, once those clarifications are in hand, the report should be revised transparently, re-running only the specific analyses that those clarifications affect, rather than the full pipeline, so that the provenance of every number remains traceable. Third, and most importantly for RQ1, a future panel study should measure tolerance and related attitudes *before* a household applies for compensation and again *after* payment is received, with a clearer set of indicators and an explicit comparison design. Only a before/after design of this kind can distinguish whether compensation genuinely lowers tolerance, whether lower-tolerance households are simply more likely to apply in the first place, or whether both are downstream of some third factor – such as prior conflict severity or grievance with the compensation process itself – that the present cross-sectional survey cannot measure.

7 Conclusion

This project set out to test whether compensation for wild boar damage raises tolerance toward wild boar among residents of two counties bordering Giant Panda National Park, and, if not, why not. Across 625 households, we find the opposite of the policy assumption: compensation participation is associated with *lower*, not higher, tolerance ($\beta = -0.109$, $p = .003$), even after adjusting for the strong confound between compensation status and county of residence ($\chi^2 = 77.1$, $p < .001$). This compensation paradox is not explained by the five mediating constructs we measured – compensation anticipation, policy support, tangible cost, intangible cost, and intangible benefit account for only 18% of the total association (RQ1) – and it does not depend on how costs and benefits interact with compensation status, since none of the three tested interaction terms is significant (RQ2). The paradox also survives extensive robustness testing: it is negative in all 8,192 alternative model specifications we estimated, and the underlying structural model is largely invariant between compensated and uncompensated residents, with one exception – conflict experience with wildlife more broadly matters more for compensated residents' tolerance than for uncompensated residents' (RQ3).

Throughout, intangible cost – residents' general evaluative attitude and emotional reaction toward wild boar – is the strongest single predictor of tolerance in every specification and every subgroup we tested, consistently outweighing compensation status itself. This points toward a broader conclusion than a narrow one about compensation policy: efforts to improve tolerance that focus only on offsetting tangible economic loss are targeting a secondary driver. The primary driver, in this data, is how residents feel about wild boar and their own capacity to cope with living alongside them – dimensions that a compensation payment alone does not address.

Two things follow for how these results should be used. First, they should be read as a robust, well-tested *association*, not as evidence that compensation payments themselves cause lower tolerance; the cross-sectional design cannot rule out that residents who were already less tolerant were also more likely to apply. Second, resolving that ambiguity – and identifying what, beyond the five mediators tested here, actually connects compensation experience to tolerance – is squarely a question for the longitudinal, before/after study design recommended in Section 6.4, not for further re-analysis of the present cross-sectional data.

Contributions

This report and the underlying analysis were produced collaboratively by the three authors, alongside project partner Mengxi Kou and supervisors Dr. Sabine Hoffmann and Marie Scherzer. Primary responsibility for each part of the analysis, the presentation, and this report was divided as follows.

- **Vaishali Iyer:** [to be completed – e.g. which sections of the data preparation, exploratory analysis, or report she led]
- **Jihao Li:** [to be completed – e.g. which sections of the data preparation, exploratory analysis, or report he led]
- **Duc-Anh Nguyen:** led the measurement-model refinement (Section 3), the primary structural (PLS-SEM) model and the RQ1/RQ2 findings (Section 4), the specification-vibration and permutation-based multi-group robustness analysis (Section 5), and drafted this report.

All three authors reviewed and approved the final presentation and this report.

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A Appendix

A.1 Glossary of abbreviations and variable codes

Table 10 collects every abbreviation and short construct code used in this report and in the accompanying analysis code, alongside its full name and a brief description. Codes are used in the running text and in tables purely for brevity and to keep cross-references to the analysis scripts unambiguous; every code is also spelled out at first use in the main text.

Table 10: Glossary of abbreviations and construct/variable codes.

Code	Full name	Description
HWC	Human–wildlife conflict	Negative interactions between humans and wildlife (crop damage, livestock loss, safety threats).
WTM	Wildlife Tolerance Model	The conceptual measurement model (Kansky et al., 2016) this questionnaire is based on.
PLS-SEM	Partial least squares structural equation modelling	The primary structural modelling technique used throughout (R package `seminr`).
EFA	Exploratory factor analysis	Data-driven check for shared correlational structure among items, without a pre-specified grouping.
CFA	Confirmatory factor analysis	Formal test of one pre-specified factor structure, with a loading and significance test per item.
PCA	Principal component analysis	Used to build the household wealth/livelihood index (Section 3.5).
HTMT	Heterotrait-monotrait ratio of correlations	Discriminant-validity diagnostic; values at or above 1.0 indicate two constructs cannot be statistically separated (Henseler et al., 2015).
AVE	Average variance extracted	Share of a reflective construct's indicator variance explained by the construct itself.
VIF	Variance inflation factor	Collinearity diagnostic for formative indicators.
VAF	Variance accounted for	Share of a total effect explained by mediation (indirect effect / total effect).
MGA	Multi-group analysis	Permutation-based test of whether structural paths differ between compensated and uncompensated residents.

Table 10 continued

Code	Full name	Description
SRMR	Standardized root mean square residual	Model-fit statistic tracked across all robustness specifications.
RQ1–RQ3	Research questions 1–3	See Section 1.
CE0	Compensation participation	Binary: applied for compensation (1) vs. did not (0). Central grouping variable throughout.
CE1, CE2	Compensation timeliness / satisfaction	Applicant-only follow-up items; skip-logic missing for non-applicants, excluded from the main model.
CA	Compensation anticipation	Formative construct: CA1 (lowest accepted %), CA2 (expected %), CA3 (insurance willingness-to-pay, log).
PS	Policy support	Formative construct: PS1, PS2 (supportive institutions, knowledge-obtained method).
TC	Tangible cost	Formative construct: TC1 (damaged crop area), log(economic loss).
IC	Intangible cost	Formative construct: IC1 (general attitude), IC2_1–IC2_3 (three-word description ratings).
IB	Intangible benefit	Single-item construct (ecological value).
PE	Prevention effectiveness	Single-item construct (average effectiveness of mitigation strategies used).
CP_WB	Conflict experience, wild boar	Formative construct: CP_WB1–CP_WB4 (frequency, duration, conflict frequency, population trend).
CP_A	Conflict experience, all species	Formative construct: CP_A1–CP_A3 (conflict-causing species count, perceived severity, duration).
InNP	Residence inside national park	Binary control variable.
Site	Survey site	Binary control variable (Baoxing vs. Qingchuan), engineered because compensation status is confounded with county (Section 2.2).
Tol1–Tol4	Tolerance items 1–4	Four Likert items measuring tolerance (Section 3.1); Tol3 is excluded from the primary construct.

A.2 Software

All analyses were run in R version 4.5.0 (R Core Team, 2025), via a `targets` reproducible pipeline (Landau, 2026). Table 11 lists, with exact version and citation, every package used at each stage of the analysis.

Table 11: R packages used, by analysis stage, with the exact version used in this analysis.

Stage	Packages (version)
Data wrangling	`dplyr` 1.2.1 (Wickham et al., 2026), `tidyr` 1.3.2 (Wickham et al., 2025), `tibble` 3.3.1 (Müller and Wickham, 2026)
Exploratory factor analysis	`psych` 2.6.5 (Revelle, 2026), `GPArotation` 2026.8-2 (Bernaards and Jennrich, 2005)
Confirmatory factor analysis	`lavaan` 0.7-2 (Rosseel, 2012), `semTools` 0.5-9 (Jorgensen et al., 2026)
Principal component analysis	`factoextra` 2.2.0 (Kassambara and Mundt, 2026)
Structural modelling (PLS-SEM, mediation, moderation, multi-group analysis)	`seminr` 2.5.0 (Ray et al., 2026)
Figures	`ggplot2` 4.0.3 (Wickham, 2016)

A.3 County confound: full contingency table

Table 12 reports the full 2 × 2 contingency table underlying the chi-squared test in Section 2.2, including expected cell counts under the null hypothesis of independence.

Table 12: County × compensation-participation contingency table. Expected counts (in parentheses) are what each cell would show if county and compensation status were independent. $\chi^2 = 77.09$, $df = 1$, $p < 2.2 \times 10^{-16}$ (Yates' continuity-corrected); odds ratio = 5.64; Cramér's $V = 0.35$.

	No compensation	With compensation	Total
Baoxing	41 (89.1)	119 (70.9)	160
Qingchuan	307 (258.9)	158 (206.1)	465
Total	348	277	625

A.4 Screening every survey variable against tolerance

Figure 10 shows the Spearman correlation of each of the 48 candidate covariates (Section 4.6) with mean tolerance (Tol1, Tol2, Tol4), one variable at a time. Ten are significant after controlling the false discovery rate at 5% (Benjamini–Hochberg; Benjamini

and Hochberg, 1995). All p-values are unadjusted for the other covariates, so the figure is a descriptive screen, not a model. Several variables are items of the same construct, so the number of significant variables is larger than the number of independent findings.

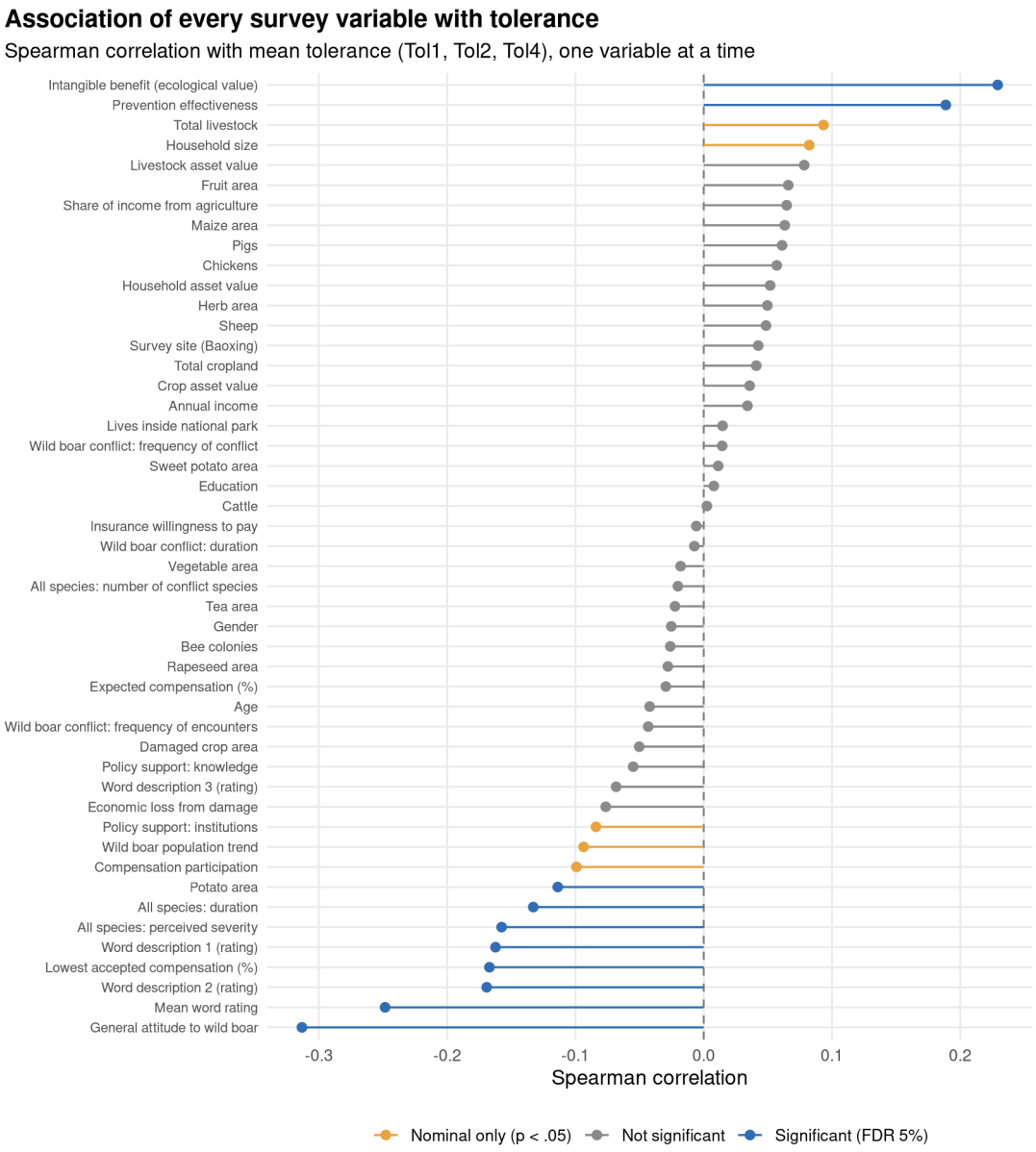


Figure 10: Spearman correlation of each survey variable with mean tolerance.

A.5 Vibration of effects on observed scores

We re-ran the vibration-of-effects exercise (Patel et al., 2015) as an independent check on observed variables: a linear regression of mean tolerance on compensation participation, with every subset of 13 candidate covariates (8,192 models). The coefficient is negative in all 8,192 models and significant at $p < .05$ in all of them, with a median of

−0.16 tolerance points (range −0.22 to −0.09; Figure 11). This is a different estimator from the PLS-based exercise in Section 5.1 (unstandardised coefficient, observed scores), which is why the percentages differ slightly.

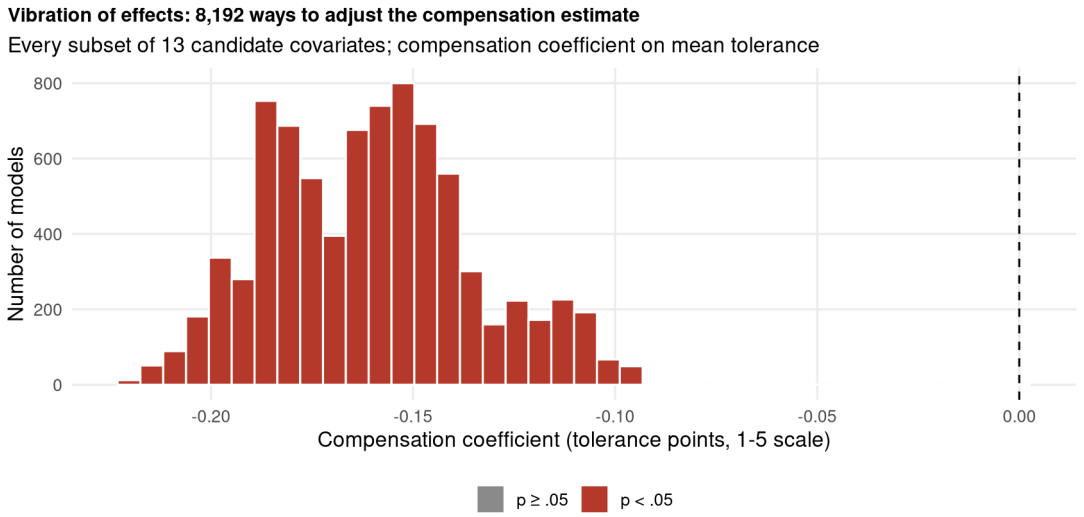


Figure 11: Compensation coefficient across all 8,192 covariate subsets.

A.6 Effect of the log transformation

Table 13: Observed-score regression of tolerance on compensation participation and the same covariates as in Section 4.3, with the money variables (income, household assets, economic loss) in four forms. Columns give the coefficients of compensation participation, intangible cost, and intangible benefit, and the p-value of household assets (robust HC3 standard errors).

Money variables	R^2	**Compens.**	**Int. cost**	**Int. benefit**	p (assets)
$\log(1+x)$ (primary)	.189	−0.168	−0.301	0.109	.026
Raw RMB	.186	−0.165	−0.294	0.109	.064
Raw, standardised	.186	−0.165	−0.294	0.109	.064
Rank	.187	−0.167	−0.303	0.107	.050

B Electronic appendix

The electronic supplement accompanying this report contains the following, organized as a single project repository:

- **This report**, as both LaTeX source (`Schreiben_Wild_boar_compensation_and_tolerance/`) and compiled PDF.
- **The presentation slides** delivered on 22 September 2026, as PDF (`Final_Presentation.pdf`).
- **Analysis code**, organized as a `targets` reproducible pipeline (`latest_analysis/scripts/`), with one numbered script per analysis stage (data preparation, exploratory data analysis, exploratory and confirmatory factor analysis, principal component analysis, the primary structural model, mediation, moderation, multi-group analysis, robustness specifications, the sensitivity checks of Section 6.2, and reporting). Running the pipeline end to end reproduces every number, table, and figure in this report from the raw survey data.
- **Output artefacts**: intermediate and final tables, figures, and fitted model objects (`latest_analysis/output/`), and a full standalone analysis write-up (`Wildboar_Compensation_Analysis_FULL_offline.html`) covering additional diagnostic detail beyond the scope of the main report.
- **The raw and cleaned survey data**, together with the project exposé supplied by the project partner (`docs/Expose_LMU_Compensation_Kou.pdf`), subject to the data protection and usage restrictions agreed with the project partner.

Code is documented with in-line comments explaining the purpose of each script and, where a choice is non-obvious, the reasoning behind it; each script can be run independently given the cleaned data object it depends on, and the full pipeline can be run end to end via `targets::tar_make()`. We aspire to, without formally certifying, the standard of the *Biometrical Journal*'s Checklist for Code Supplements.

Declaration of authorship

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